

1 Theoretical Framework

This thesis treats LLM-based evaluators as measurement instruments that map an input (e.g., marketing copy text) to a score that represents an underlying (subjective) construct (e.g., "Quality"). The goal is to align the LLM evaluation with human evaluation.

The construct of interest can be evaluated directly, in which case it is considered holistically. However, a fundamental challenge in evaluating subjective attributes is that they are not singular, physical properties like temperature or length, but high-dimensional constructs. To address this, we propose a methodology inspired by *construct representation*, treating the task of evaluation as a cognitive process which can be decomposed into sub-processes [6, 11]. We hypothesize that by explicitly modelling the criteria humans implicitly use to reach a judgment, an LLM can yield a better measurement of the construct compared to a holistic evaluation.

A similar measurement approach where a complex construct is decomposed is taken in Li and Sun [3] and Wang et al. [10]. Li and Sun [3] note that it is difficult to align general attributes such as "usefulness" and "harmlessness" with human evaluations, so they decompose them into more specific criteria. Wang et al. [10] find that an "analytic scoring" approach outperforms holistic scoring, where they supply the LLM with Rubrics containing multi-dimensional assessment criteria on which it bases its evaluation.

However, these studies typically treat decomposition as a pragmatic scoring or training design choice, rather than as an explicit construct-specification problem. As a result, it remains unclear how to adequately define the evaluation criteria, how the decomposed scores should be interpreted, and which criteria for validity follow. Validity in psychometrics concerns whether an instrument measures what it is supposed to measure, and serves as support for interpretations and actions based on the measurements. It is established through both theoretical rationales as well as empirical evidence [7]. By making the measurement model explicit, we can define the theoretical requirements for meaningful measurements and derive an empirical validation strategy that matches the construct specification.

Moreover, common choices in the operationalization of LLM-based evaluators can introduce construct-irrelevant variance. In particular, placing multiple criteria in a single prompt makes scores sensitive to criterion order ("position bias"; see Shi et al. [8]), and requesting multiple scores in a single generation mechanically couples later scores to earlier outputs via autoregression. We therefore measure each criterion in isolation, consistent with Bang et al. [1]. Our contribution is to formalize why this operationalization improves measurement validity.

Finally, the validation methodologies in Li and Sun [3] and Wang et al. [10] require improvement. Firstly, they establish *convergent validity* by comparing LLM ratings to human ratings on identical analytic scales. This approach demonstrates that an LLM can mimic human grading on specific sub-dimensions, but fails to prove that the decomposition strategy itself yields a superior measurement of the overall quality compared to a holistic evaluation. The theoretical

foundation from Section 1.1 allows us to specify a model to directly assess this. Furthermore, neither paper has access to a dataset which includes behavioral outcomes, limiting their validation to human proxy labels. Demonstrating functional utility is especially important considering the intended use of the evaluator on the Barter platform. We therefore extend validation by assessing *predictive validity* against a downstream behavioral KPI.

Accordingly, the structure of this chapter is as follows.

- **Theoretical definition:** We explain the two causal models that define the relationship between a composite construct ("latent variable") and its constituents ("indicators"): the *reflective* and the *formative* model.
- **Computational operationalization:** We formalize the independent measurement of these constituents by defining the mathematical mapping of dimensions to logit-based scores.
- **Empirical validation:** We explain how we assess measurement validity through the psychometric concept of *construct validity*. We assess construct validity through *convergent validity* (alignment with human holistic labels) and *predictive validity* (prediction of a behavioral KPI).

1.1 Theoretical Definition: Latent Variable Models

Latent variable models define a relationship between a target construct and its indicators. Two model types are distinguished: formative and reflective [2]. They differ in their assumptions about the direction of causality.

In formative models, the latent variable is assumed to be a composite construct, caused by the indicators ("causal indicators"). The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and prestige [5]. Formally, for a set of causal indicators indexed by $i = 1, 2, \dots, n$, we assume

$$\eta = \sum_{i=1}^n \gamma_i x_i + \zeta \quad (1)$$

where η is a latent construct, γ_i a formative weight, x_i a manifest variable, ζ the disturbance term.

In reflective models, the latent variable causes the manifest variables ("effect indicators"). For example, following a "g-factor" interpretation of the latent variable intelligence [9], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any effect indicator i , we assume

$$x_i = \lambda_i \eta + \varepsilon_i \quad (2)$$

where x_i is manifest variable, λ_i a factor loading, η the latent construct, and ε_i the measurement error.

Exemplars in LLM-based measurement. Decomposition already appears in LLM-based evaluation, but with different implicit measurement stances. Wang et al. [10] study LLM scoring in large-scale writing assessment using psychometric reliability frameworks, consistent with treating rubric dimensions as manifestations of a common underlying writing ability (reflective stance). Li and Sun [3] operationalize broad evaluation targets through batteries of criteria organized by task category, emphasizing domain coverage, which is consistent with treating evaluation as a composite defined by its constituents (formative stance).

Implications. These contrasting stances illustrate that the assumed causal model has downstream consequences for both measurement design and validation. Under a reflective specification, indicators are treated as effects of a common underlying attribute; accordingly, strengthening the measurement design (e.g., additional tasks/raters, well-functioning indicators) improves the precision of the latent estimate without redefining the construct, and internal-structure evidence (e.g., factor structure and reliability/generalizability) is a primary diagnostic of measurement quality. Under a formative specification, the construct is defined by its constituents; indicators are not expected to be interchangeable or strongly intercorrelated, and adding or omitting an indicator changes the construct’s meaning. Measurement quality therefore hinges on defensible construct specification and content coverage and accurate measurement of each constituent, while validation relies primarily on external relations (e.g., alignment with alternative measures of the target attribute and prediction of relevant criteria).

(relate formative to "content validity", which would be the theoretical requirement for meaningful measurement)

Construct specification (quality-for-purpose). In this thesis, “marketing copy quality” is treated as a quality-for-purpose construct: the attribute is defined by the requirements of its intended use in the Barter setting (i.e., what stakeholders need the copy to achieve). Consequently, the construct is not assumed to be a single underlying trait that causally manifests in multiple observed ratings, but a composite defined by a set of constituent criteria (e.g., clarity, persuasiveness, and compliance with offer details) selected to cover the relevant content domain. This motivates a formative specification: the indicator set constitutes the construct definition, so modifying the set changes what “quality” means. The role of empirical analysis is therefore not to discover a latent factor behind the indicators as in a reflective model, but to test whether the specified composite yields valid measurements with respect to human judgments and downstream behavioral outcomes.

1.2 Operationalization: LLM as a Measurement Instrument

Following the formative specification defined in Section 1.1, we must define a computational process that measures each indicator x_i (corresponding to criterion c_i) independently. We operationalize the LLM as a probabilistic measurement instrument. This section formalizes the mapping from input text to

score and demonstrates why standard generation techniques introduce construct-irrelevant variance through positional and autoregressive dependencies.

Input Specification (Encoding) Formally, an LLM maps an input token sequence $\mathbf{u}$ to a distribution of next-token logits $\mathbf{z}(\mathbf{u}) \in \mathbb{R}^{|\mathcal{V}|}$ over a vocabulary $\mathcal{V}$. The conditional probability of a token v is given by the softmax function:

$$P(v \mid \mathbf{u}) = \frac{\exp(z_v(\mathbf{u}))}{\sum_{w \in \mathcal{V}} \exp(z_w(\mathbf{u}))}. \quad (3)$$

In standard evaluation pipelines, multiple criteria are often concatenated into a single input sequence $\mathbf{U}(d) = [s; c_1; \dots; c_I; d]$. However, due to the self-attention mechanism, the representation of any specific criterion c_i becomes a function of its position within $\mathbf{U}$. This introduces position bias, where the measurement is sensitive to the order of instructions rather than the content of the document d [8].

To ensure the measurement of any indicator is invariant to the presence or order of other indicators, we define the input strictly as the triplet of task instruction s , single criterion c_i , and document d . For each criterion i , we construct an independent input sequence $\mathbf{u}_i$:

$$\mathbf{u}_i(d) = [s; c_i; d]. \quad (4)$$

This ensures that the logit vector $\mathbf{z}(\mathbf{u}_i)$ depends exclusively on the instructions, the definition of c_i , and the document content.

Score Extraction (Decoding) Standard approaches often require the model to generate a sequence of scores $\mathbf{y} = (y_1, \dots, y_I)$ for the batched criteria. This relies on autoregressive decoding, where the probability of a score y_t is conditioned on the previously generated scores $y_{<t}$:

$$P(\mathbf{y} \mid \mathbf{U}) = \prod_{t=1}^J P(y_t \mid \mathbf{U}, y_{<t}). \quad (5)$$

where t denotes the position in the token sequence. This formulation introduces two sources of construct-irrelevant variance. First, it creates *autoregressive coupling*: the measurement of the t -th indicator is mathematically dependent on the values assigned to previous indicators, violating the requirement that a measurement should depend only on the attribute being measured. Second, the generation of a multi-token sequence relies on stochastic sampling strategies (e.g., nucleus sampling) and decoding hyperparameters (e.g., temperature), introducing stochasticity that degrades measurement consistency.

To eliminate these dependencies, we restrict the generation horizon to a single token. We treat the evaluation of criterion c_i as an independent classification task and extract the score y_i by directly analyzing the next-token distribution conditioned on $\mathbf{u}_i(d)$:

$$y_j \sim P(\cdot \mid \mathbf{u}_i(d)). \quad (6)$$

By constraining the "generation" to the first step, we ensure that $P(y_i)$ is conditioned strictly on the input $\mathbf{u}_i(d)$ and never on prior outputs. Furthermore, this approach grants direct access to the logits $\mathbf{z}(\mathbf{u}_i)$, allowing us to bypass the sampling step entirely and derive deterministic, reproducible measurements invariant to decoding strategies.

1.3 Empirical Validation: Construct Validity

In order to determine whether the operationalization aligns with the theoretical constructs, we adopt a *construct validity* approach. Recall that validity is established through both theoretical rationales and empirical evidence [7]. Where Section 1.1 provided the former, this section provides the latter.

The motivation for adopting a construct validity approach is the same for psychology as it is for our evaluations. Unlike objectively measurable properties like "temperature" or "grammatical correctness", the latent (unobservable) nature of constructs means there is no observable "true" value to validate against. Instead, establishing construct validity requires accumulating evidence from multiple sources to demonstrate the appropriateness of the interpretations and their intended use [7]. In this thesis, we build this evidence through two types of construct validity: *Convergent validity* (alignment with human expert judgment) and *predictive validity* (alignment with downstream behavioral outcomes).

Convergent Validity. To assess convergent validity, we treat human expert ratings as a reference measure of the target construct. Let $y^H(d)$ denote a human holistic rating for document d . We assess validity by modeling the relationship

$$y^H(d) = g(\mathbf{x}(d)) + \epsilon(d). \quad (7)$$

This formulation directly implements the formative specification defined in Equation 1 (Section 1.1). By treating the human judgment y^H as a proxy for the latent construct η , we can empirically estimate the weights γ that best predict the holistic assessment from the independent indicators.

We evaluate whether the LLM-derived indicators $\mathbf{x}(d)$ explain systematic variation in $y^H(d)$ on held-out data. Importantly, this evaluates the *construct-level* utility of decomposition: rather than only testing whether an LLM can imitate human ratings on the indicator level as in Li and Sun [3] and Wang et al. [10], we test whether the indicator measurements can be combined into a superior predictor of the overall expert judgment.

We compare this against a holistic baseline $x^{\text{hol}}(d)$ obtained by prompting the LLM for a single overall "Quality" score, so we can compare

$$y^H(d) \approx g(\mathbf{x}(d)) \quad \text{vs.} \quad y^H(d) \approx h(x^{\text{hol}}(d)). \quad (8)$$

If formative decomposition improves measurement, the composite model should achieve lower predictive error than the holistic baseline.

Predictive validity. Predictive validity assesses whether a measure exhibits the theoretically expected relationships with an external measure [6]. In our "quality-for-purpose" framework, this corresponds to functional utility: whether the measurements can predict the downstream outcomes the copy is expected to drive.

Let $y^{\text{KPI}}(d)$ denote a behavioral outcome associated with document d (e.g., user engagement or conversion on a platform). We test whether the vector of criterion measurements predicts this outcome:

$$y^{\text{KPI}}(d) = r(\mathbf{x}(d)) + \eta(d), \quad (9)$$

We again compare this performance to the holistic baseline $x^{\text{hol}}(d)$. Evidence that $\mathbf{x}(d)$ is a stronger predictor of $y^{\text{KPI}}(d)$ supports the construct-validity claim that the measurements are appropriate for their intended use, i.e., guiding diagnosis, validation, and improvement of marketing copy in deployment settings.

Together, these two forms of validity provide evidence that our operationalization yields meaningful and usable measurements of the construct under the formative specification. While additional measures of construct validity can be considered (see: Lin [4]), convergent validity and predictive validity target the two principal requirements in our setting: alignment with human evaluations, and functional utility in deployment.

Bibliography

- [1] G. Bang, D. Kim, and M. Min. Transforming user defined criteria into explainable indicators with an integrated LLM AHP system. URL <http://arxiv.org/abs/2601.05267>.
- [2] K. Bollen and R. Lennox. Conventional wisdom on measurement: A structural equation perspective. *Psychological bulletin*, 110(2):305, 1991.
- [3] Y. Li and Y. Sun. A fine-grained evaluation framework for language models: Combining pointwise grading and pairwise comparison. 63(1): 104270. ISSN 0306-4573. <https://doi.org/10.1016/j.ipm.2025.104270>. URL <https://www.sciencedirect.com/science/article/pii/S0306457325002110>.
- [4] Z. Lin. A validity-guided workflow for robust large language model research in psychology. URL <http://arxiv.org/abs/2507.04491>. Read_Status: In Progress Read_Status_Date: 2025-10-14T14:31:31.251Z.
- [5] S. McQuitty and M. Wolf. Structural equation modeling: A practical introduction. *Journal of African Business*, 14(1):58–69, 2013.
- [6] S. Messick. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. 1994(2):i-28. ISSN 2330-8516. <https://doi.org/10.1002/j.2333-8504.1994.tb01618.x>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/j.2333-8504.1994.tb01618.x>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.2333-8504.1994.tb01618.x> Read_Status: New Read_Status_Date: 2025-10-14T12:42:52.831Z.
- [7] S. Messick. Meaning and values in test validation: The science and ethics of assessment. *Educational researcher*, 18(2):5–11, 1989.
- [8] L. Shi, C. Ma, W. Liang, X. Diao, W. Ma, and S. Vosoughi. Judging the judges: A systematic study of position bias in LLM-as-a-judge. URL <http://arxiv.org/abs/2406.07791>.
- [9] C. Spearman. "general intelligence" objectively determined and measured. 1961.
- [10] Y. Wang, J. Huang, L. Du, Y. Guo, Y. Liu, and R. Wang. Evaluating large language models as raters in large-scale writing assessments: A psychometric framework for reliability and validity. *Computers and Education: Artificial Intelligence*, page 100481, 2025.
- [11] S. E. Whitely. Construct validity: Construct representation versus nomothetic span. *Psychological bulletin*, 93(1):179, 1983.

A Prompt rendering via `apply_chat_template`

We format each prompt using Hugging Face’s chat template with `add_generation_prompt=True`, and then append an explicit assistant prefix (shown below). The rendered template is passed to the model, which computes the next-token probabilities after the assistant prefix. The model input is the concatenation of:

(i) system message, (ii) user message, (iii) assistant header inserted by the template, (iv) `assistant_prefix`.

```
<System role header as produced by tokenizer>
Instruction: Rate the Relevance of the Answer provided for
            the Question
on a scale of 1 to 4. Use the following rubric to determine
            the score:

Relevance: Does the answer discuss the specific topic and
            entities requested?

1 = Irrelevant: Discusses a completely different topic (e.g.,
            "baby delivery"
instead of "respirators").

2 = Topic Mismatch: Discusses a related category but the
            wrong specific
entity (e.g., "Work Visa" instead of "Pandemic Visa").

3 = Broadly Relevant: Discusses the correct topic but
            includes significant
tangential or unrelated information.

4 = Precise: Focuses exclusively on the specific entity and
            topic requested
in the question.
</System role header>

<User role header as produced by tokenizer>
Question:
{question}

Answer:
{answer}
</User role header>

<Assistant role header inserted by add_generation_prompt=True
>
Rating:
```


B Sample data

B.1 FeedbackQA examples

Example F01

Question: Is it still possible to access medical appointments and medications through my doctor?

Answer: You can access bulk-billed appointments with your doctor, nurse or mental health professional via phone or videoconference. If you would like to arrange a telehealth appointment, contact your doctor. Accessing medicines while staying at home. Your doctor may be able to arrange to have your prescriptions sent to your pharmacy directly so you can stay at home. For more information on whether you can access this service please call your pharmacy and speak to them about the best way to arrange this.

Rating 1: 4

Feedback 1: Gives information on how patients can access appointments with their doctors during the pandemic.

Rating 2: 3

Feedback 2: This answer relates to accessing medical appointments and medications, but does not give much information.

Table 1. Sampled FeedbackQA items (IDs and labels), and Qwen scores for Relevance, Completeness, Directness, and Holistic criteria

ID	Human label 1	Human label 2	Rel.	Comp.	Dir.	Holistic
F01	4		3	4	3	3

B.2 Barter Deals examples

Example B01

Deal text: We'd love a video capturing the experience of visiting our store in the heart of Amsterdam—from the vibe of the city and the walk over, to exploring the shop itself. While you're there, pick your favorite bag from the collection and make it part of the story!

Keep it authentic, stylish, and feel free to show the charm of the surroundings too!

Applications after 7 days: 39

Product category: Activities

Example B02

Deal text: Choose 2 coffees and pastries you like:)

Applications after 7 days: 6

Product category: Food